

The block diagram illustrates the hardware architecture of the proposed system. It features an FPGA (Field-Programmable Gate Array) which houses a CORE containing a DEBUG MODULE. The system is interfaced with several external components and signals:

- Inputs/Outputs:** SW8, SW7, and SW4 are switches providing inputs. A PC is connected via JTAG (JTAG interface) for TDI, TMS, TCK, and TDO signals. An LED display shows the output.
- Internal Components:** The CORE includes a DEBUG MODULE. Main Memory is connected to the CORE. A multiplexer (MUX) selects between R17 and ext_rdata to drive the LED display.
- Signals:** RESET, Ext_addr, R17, ext_rdata, Debug Status, and TDI/TMS/TCK/TDO are key signals in the system.

- 8KB, 2 Way Caches
- 5-Stage Pipelined
- Memory Protection Unit
- Memory Management Unit
- Branch Prediction Unit
- Machine Mode CSRs
- Software and Timer Interrupt
- Exception and Trap Handling
- Debug Support